GoreeCloud Search Fusion Engine - Development integration contract

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Stage: Development-only experimental source on the GoreeCloud Search development branch. **Not** connected to the deployed website, live external providers, or a public API. An offline, opt-in versioned fusion path is now included in the *candidate* native Linux .pyz; no new service was deployed.

native_search/fusion.py contains the first independent GoreeCloud-owned result-fusion primitive. It accepts immutable SourceResult cards for web, images, videos and news, and emits FusedResult objects. Its standard-library implementation performs no I/O or background requests.

native_search/dispatch.py adds the **Development-only adapter-dispatch contract**. It accepts a bounded query and a finite set of named asynchronous provider adapters, checks their explicit per-category ProviderRights, runs only admitted adapters with cooperative deadlines and a maximum concurrency limit, and combines their normalized cards through the existing ranker. It returns a versioned FusionResponse with one fused result list and query-free per-provider health states (blocked, ok, empty, timeout, failed). It neither ships provider adapters nor authorizes provider use.

native_search/api.py exposes POST /v1/fusion on the Linux-native service's existing 127.0.0.1 listener. The default **Web** and **Images** categories now fuse GoreeCloud's own local SQLite FTS5 documents and operator-approved image metadata through sealed goreecloud_index and goreecloud_images adapters. No external provider is configured or contacted by default. The Images route returns an honest HTTP 200 empty response if the owned image index has no matching records; that does not imply external image coverage. Unconfigured videos and news still return HTTP 503. A developer must explicitly start the service with --fusion-fixture /absolute/path/to/synthetic.json to admit strictly bounded, local synthetic cards for an offline preview. The response combines source cards in one category and reports only bounded, query-free result cards and provider states; it is not live metasearch.

v1 offline dispatcher boundaries

- Rights default to **deny**. Fetch, fusion and display must each be explicitly allowed and accompanied by a nonempty reference to reviewed upstream terms; category membership is also explicit. rights_basis defaults to unverified and cannot admit. synthetic_fixture admits only with the explicit allow_synthetic_fixtures=True testing switch, **off by default**; contract_reviewed is an operator assertion requiring independent documentary confirmation before configuration. No shipped provider has this assertion. A code flag is not independent legal permission, nor is a synthetic fixture rights grant permission for Google, Bing, Brave, or any other real upstream.
- First-party owned-index admission is separate from the external-rights gate. For Web, source goreecloud_index, evidence goreecloud-local-sqlite-fts5, exact OwnedIndexAdapter type and owned_index rights must match. For Images, the respective requirements are goreecloud_images, goreecloud-local-sqlite-images-fts5, exact OwnedImageAdapter type and owned_index rights. Each adapter is sealed to its one category; neither performs outbound fetches. An external adapter cannot borrow either reserved identity, and a self-declared owned_index token cannot authorize network access.
- The owned Images database stores *metadata only*: approved page URL, original image URL, title, description and nonempty operator rights-evidence reference. Page and media links must be canonical public HTTPS URLs without credentials, queries, fragments or private-network targets. A bounded, version-one local

manifest (schema_version: 1, images list, up to 100 records) can be imported with python -m native_search import-images-json --db DB --input /absolute/operator-approved-images.json. This operator-only command is in the **source checkout**, not a packaged .pyz ingestion interface. Media files are never downloaded by this operation. Operators must independently verify image-display rights and can remove stored image metadata promptly; no outside feed, crawling permission or third-party image license is implied.

- Up to eight named providers may be registered; each can return at most 40 SourceResult cards, with at most four concurrent calls **within one invocation**. The current Linux-native HTTPServer has one synchronous request worker and therefore limits fusion calls to one at a time in that *specific* Development process. It also bounds idle accepted client connections to five seconds. The dispatcher alone does not impose cross-process or multiworker concurrency budgets. A source receives the query only if admitted; inaccessible or denied sources are not called.
- The default cooperative per-provider deadline is three seconds (maximum ten), starting **after its semaphore slot is acquired** so a queued source is not timed out without a chance to execute. Real adapters **must** be asynchronous, honor cancellation, enforce their own socket/client deadline, upstream rate budget and egress policy, and never spawn unbounded blocking work. This module is not a process-level kill switch for a malicious or stuck third-party connector.
- Invalid adapter shape, mixed source identity, invalid ranks and unpublishable/unsafe result URLs fail that **one provider** with zero accepted cards instead of treating filtered cards as healthy results. This preserves other admitted providers' valid results. Source errors are counted, not logged or reflected with query-bearing exception messages; no hidden retries, substitute providers or user-query cache are implemented.
- FusionResponse has schema_version=1, category, a unified results tuple, source providers outcome tuples, and overall status (ok, partial, empty, unavailable). partial covers degraded or blocked configured providers; empty indicates permitted sources responded but produced no publishable results. Only the isolated native loopback adapter exposes this as Development JSON. This is **not** a public route or a promise of user-facing production behavior.
- The native fusion service may read the owner's local FTS5 indexes for Web and approved Images; synthetic provider fixtures remain offline and opt-in. The currently deployed SearXNG-derived Development site, Caddy ingress, public host and source provider configuration are unchanged; a new unshipped .pyz candidate contains the native offline fusion route.

v0 deterministic behavior

- Every card records a validated lowercase source identifier, category, one-based position, HTTP(S) destination, title and optional description or image media URL; invalid destinations/cards are discarded. Category mismatch, malformed source identity, invalid rank or oversized input fails closed.
- Web and News use conservative destination URL equality, stripping fragments, default ports and listed tracking parameters. Videos also recognize a narrowly validated identical YouTube clip across watch, youtu.be, Shorts, live and embed URLs. Ambiguous IDs, nonstandard ports, mirrored hosts, syndication copies and separate news stories are **not** automatically merged. The original display destination is preserved.
- Images deduplicate by a validated original/media URL, never just the page hosting the asset. Images without media identity remain separate until richer evidence exists. Perceptual hashing is not implemented.
- A duplicate receives at most one vote per distinct source; if a single source returns that URL more than once, its best position wins. A score sums 1 / (60 + position) across its source votes (reciprocal rank fusion). Stable encounter order breaks exact-score ties. An optional *validated* search query adds deterministic whole-token title, description and exact-phrase credit, bounded to **20% of one source's rank-one vote**. The dispatcher validates at most 16 distinct query terms before contacting providers and applies the signal only at the final fusion step. It cannot invent additional provider votes, and omitted-query calls preserve the original reciprocal-rank scores and order.
- Source/rank evidence is internal to the model, not a provider-selection requirement in the product UI. Exposing or hiding attribution is subject to each approved provider agreement.

- Inputs are bounded to 1,000 cards, output to 100; the pure fusion primitive has no network I/O, logging, HTTP handler, SQLite table, cache or provider SDK. The optional query is used solely for in-memory ranking. Title (512 characters), description (2,000) and URL (4,096) fields are separately bounded before inclusion.

Integration acceptance still required

1. **License gate:** Only explicit, source-specific allowed usages can reach a provider adapter or populate the production fuse input. A *technically accessible* upstream HTML page is not contractual permission. Brave's derivative/caching restrictions and Google/Bing API lifecycle changes must be resolved before their native fusion activation.
2. **Adapter contract:** Map provider-specific payloads into typed vertical cards, validate redirects/media and language/locale, enforce per-provider limits, budget, timeout, network egress and content rights. Partial failures must not be disguised as an empty index.
3. **Orchestrator:** The offline cooperative dispatch contract exists, but it still needs a deployment-qualified async transport, signed and reviewed source-rights configuration, cross-request worker/request budget, upstream rate-limit/backoff policy and cancellation under real outages. Keep no invisible fallback source, expose degraded-state telemetry without raw user queries, and explain privacy implications of automatic provider fanout. Prefer POST for private Search to avoid query URLs.
4. **Ranking acceptance:** A bounded lexical query-overlap signal is now Development-only. Compare it against pinned synthetic and separately authorized real relevance judgments; add recency, quality/spam, domain diversity, language and visual/media similarity only after benchmark evidence. Lexical overlap alone is not semantic relevance or a safety signal.
5. **Runtime:** A real loopback-owned `POST /v1/fusion` exists for local-index Web/Images results and optional synthetic Development data. It is not the deployed public Search service. Provide load, stalled-client, recovery, rate limiting, security, authentication, privacy, observability and rollback acceptance on the exact candidate before any public or supported-native-OS release claim.

Offline tests: `python3 -m unittest discover -s tests/native -p test_fusion.py -v`. Dispatch tests: `python3 -m unittest discover -s tests/native -p test_dispatch.py -v`. The candidate native Linux `.pyz` allowlist now includes the owned fusion, dispatch, sealed index-provider and synthetic-fixture modules. Local package tests and source-revision checks do not substitute for install, systemd lifecycle, signed-distribution, upgrade, real-provider, or live ingress acceptance.

Current integration constraints: Bing Search APIs retired in August 2025; Google Custom Search JSON API is closed to new customers and scheduled to end January 1, 2027; Brave Search API result storage, derivatives and redistribution require an agreement-specific legal review. None is enabled by this change. Official lifecycle/terms references: https://learn.microsoft.com/en-us/lifecycle/announcements/bing-search-api-retirement; https://developers.google.com/custom-search/v1/overview; https://api-dashboard.search.brave.com/documentation/resources/terms-of-service.